AI-AUGMENTED METHODS FOR INTERPRETABLE AND EFFICIENT MODELING IN MODERN ASTRONOMY

25-26J-108

Project Proposal Report

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Department of Computer Science

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August 2025

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Dilmani H.V.N.	IT22122032	Dilmani.

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

.... 31/08/2025

Signature of the Supervisor: Date:

DEDICATION

This work is dedicated to our supervisor, Mr. Samadhi Chathuranga; our co-supervisor, Dr. Kapila Dissanayake; our external supervisor, Mr. Banuka Dimuthu; and to all the friends and family whose guidance, encouragement, and support have been invaluable throughout this journey.

ACKNOWLEDGEMENT

I would like to extend my heartfelt gratitude to our supervisor, Mr. Samadhi Chathuranga, our co supervisor, Dr. Kapila Dissanayake, and our external supervisor for their generous time, guidance, and support throughout this journey. Your encouragement and expertise have been invaluable in shaping my work and helping me overcome challenges.

ABSTRACT

In contemporary astronomy, data-driven methods such as deep learning, self-supervised learning, and explainable AI have matured sufficiently to enable both accurate and interpretable scientific modeling. This study applies attention augmented recurrent neural networks (RNNs) to model quasar luminosity variability [1], [2], aiming to capture complex, nonlinear patterns in their brightness over time. Quasars—powered by supermassive black holes—exhibit irregular light-curve behavior that is difficult to analyze using traditional methods. By integrating attention mechanisms into RNNs, our approach highlights the most informative time segments in quasar light curves, improving both predictive accuracy and interpretability. We train and evaluate the model on extensive time-series datasets, comparing its performance to standard sequence models [3], [4]. The attention enhanced RNN not only demonstrates superior modeling of quasar variability but also offers insight into the temporal drivers of quasar brightness changes. These findings underscore the potential of attention-based sequence learning as a powerful, interpretable tool for understanding active galactic nuclei dynamics and other astrophysical time-series phenomena.

Key Words - Quasar variability, time-series analysis, recurrent neural networks, attention mechanism

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LIST OF ABBREVATIONS

- **AGN** - Active Galactic Nucleus / Active Galactic Nuclei
- **AI** – Artificial Intelligence
- **RNN** - Recurrent Neural Network
- **DRW** - Damped Random Walk
- **LSTM** - Long Short-Term Memory
- **PINN** - Physics-Informed Neural Network
- **u, g, r, i, z** - Photometric filter bands in SDSS (ultraviolet, green-blue, red, near-infrared, deep infrared

1. INTRODUCTION

1.1 Background and Literature Survey

Quasars are extremely luminous objects powered by supermassive black holes at the centers of galaxies. Their light output varies irregularly over time due to complex physical processes such as accretion disk instabilities, relativistic effects, and variations in matter inflow. Studying quasar luminosity variability is important for understanding black hole growth, accretion physics, and the structure of active galactic nuclei (AGN).

Traditional time-series analysis methods, such as Fourier transforms and autoregressive models, often struggle to capture the highly non-linear and non-stationary nature of quasar light curves. In recent years, deep learning techniques—particularly recurrent neural networks (RNNs) and their variants—have shown strong capabilities in modeling sequential and temporal data [1], [3].

The addition of attention mechanisms has further improved sequence modeling by allowing networks to focus on the most relevant parts of the input when making predictions [2]. This is particularly valuable for quasar studies, where significant events may be localized in specific time intervals of the light curve.

Previous studies have successfully applied deep learning to astronomical phenomena such as supernova classification, galaxy morphology analysis, and gravitational wave detection [4]. However, the application of RNNs to quasar variability remains relatively unexplored. This research aims to bridge that gap by integrating domain knowledge (basic physical constraints) with modern sequence learning techniques to improve both prediction accuracy and interpretability.

Below figure 1 and figure 2 results obtained from traditional variability modeling techniques.

A multiband quasar light curve derived from standard photometric monitoring, showing brightness variations across u, g, r, i, z filters. It helps reveal physical processes, like black hole accretion activity, by comparing brightness changes across colors. By comparing how bright a quasar looks through each “color,” scientists can learn about its temperature, composition, and how it changes over time.

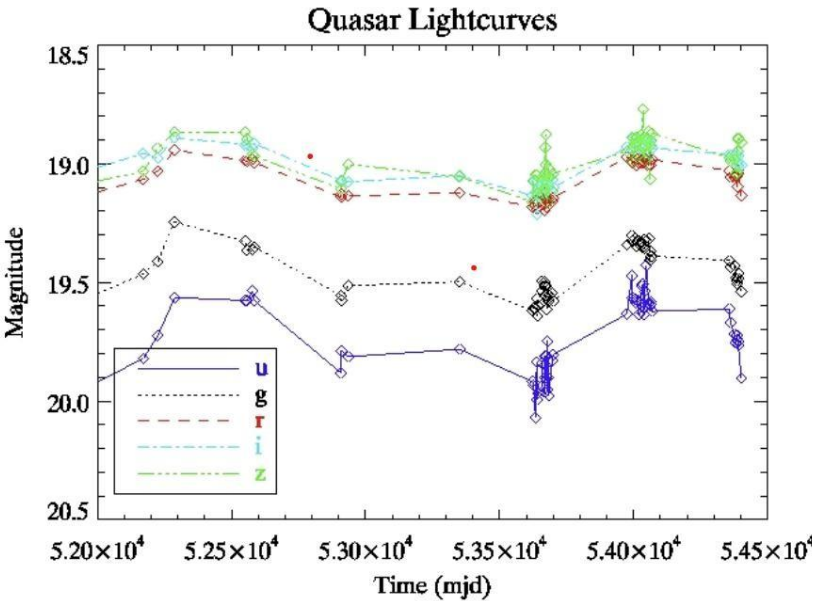

Figure 1 - From University of Washington astronomy resource:faculty.washington.edu – Quasar Variability Project

Figure 2 is a graph showing a quasar's actual brightness changes over time (dots with error bars) and a smooth curve from a damped random walk (DRW) model trying to match them.The point is: the DRW curve doesn't fully capture the quasar's complicated ups and downs, showing why more advanced methods (like your RNN with attention) are needed.

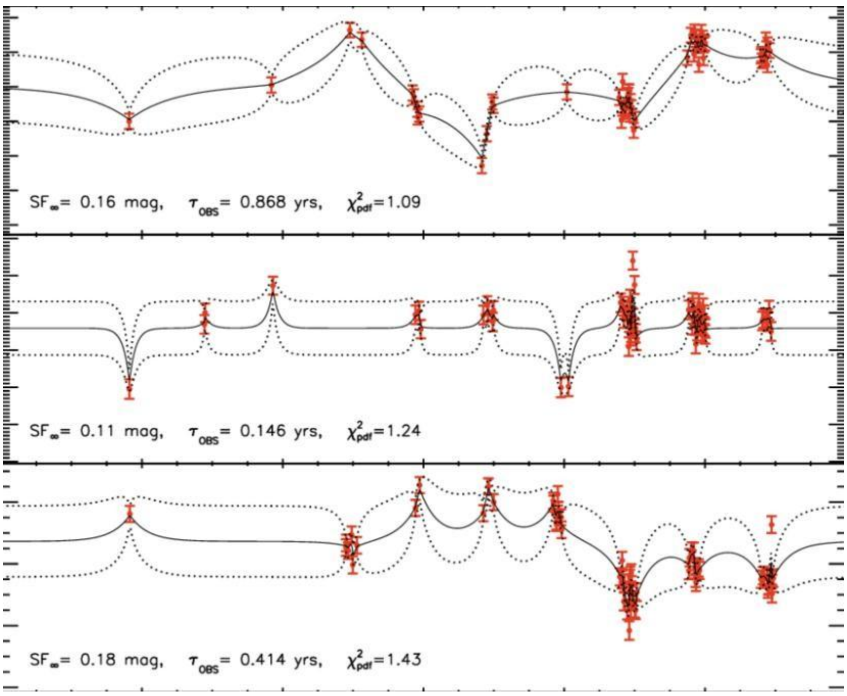

Figure 2 - quasar variability modeling

1.1.1 Use of Physics-Informed AI in Astronomy

In quasar variability studies, physics-informed AI integrates established astrophysical principles such as light-travel time effects and energy conservation directly into machine learning models [5]. By embedding these constraints, the model's predictions remain consistent with known quasar physics while still leveraging the pattern-recognition power of deep learning. This approach reduces overfitting, improves generalization to unseen light curves, and enhances the physical interpretability of model outputs, making it especially valuable for analyzing irregular and complex brightness variations in active galactic nuclei.

1.1.2 Interpretable AI in Astronomy

RNNs with attention can show which parts of a quasar's light curve matter most for predictions, making the results easier to understand [2]. These models can highlight the most relevant time segments influencing predictions, offering a degree of interpretability. However, integrating attention-based interpretability with physics-informed constraints is still uncommon, even though it would ensure that model insights remain both data-driven and physically consistent an approach especially promising for quasar variability analysis.

1.1.3 Applying Machine Learning to Understand Time-Series Analysis of Quasars

Quasars, among the most luminous objects in the universe, display highly irregular variability in their brightness over time due to complex astrophysical processes occurring near supermassive black holes. Modeling this variability poses significant challenges because quasar light curves are non-linear, non-stationary, and often noisy. Traditional time-series methods, such as Fourier analysis or autoregressive models, frequently struggle to capture these dynamics. In recent years, deep learning—particularly sequence modeling architectures like Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Gated Recurrent Units (GRUs)has emerged as a powerful alternative for handling such complex temporal data [1], [3].

In quasar variability studies, physics-informed AI integrates established astrophysical principles—such as light-travel time effects and energy conservation—directly into machine learning models. By embedding these constraints, the model's predictions remain consistent with known quasar physics while still leveraging the pattern-recognition power of deep learning. This approach reduces overfitting, improves generalization to unseen light curves, and enhances the physical interpretability of model outputs, making it especially valuable for analyzing irregular and complex brightness variations in active galactic nuclei (AGN). The synergy between data-driven modeling and physics-based constraints represents a promising path forward for extracting both accurate and scientifically meaningful insights from astronomical time-series data.

1.2 Research Gap

Current approaches to quasar property estimation often rely on traditional spectral analysis methods, such as manual or semi-automated fitting of emission and absorption lines. While precise, these methods can be time-consuming, highly sensitive to noise, and limited in their ability to handle complex or blended features. Many machine learning models applied to quasar studies focus primarily on pattern recognition without incorporating astrophysical constraints, which reduces the physical reliability and interpretability of their predictions [1], [3]. Furthermore, there is limited integration between deep learning–based variability modeling and established spectral or variability analysis techniques, resulting in fewer opportunities to cross-validate results for scientific consistency. Even when advanced models such as RNNs with attention are used, they often do not clearly reveal which parts of the light curve most strongly influence predictions, making it difficult for astronomers to verify the reasoning behind the outputs [2]. Finally, much of the existing work depends on simulated datasets rather than extensive real observational data, limiting the applicability of findings to actual astronomical observations. This leaves a clear gap for a modeling framework that is both physically consistent and transparent, while grounded in real quasar light curve measurements. Observations could yield more realistic and impactful results.

1.3 Research Problem

Quasars exhibit highly irregular and complex brightness variations driven by physical processes near supermassive black holes. Traditional methods such as spectral line fitting and basic time-series struggle to fully capture these patterns, especially when dealing with noisy, incomplete, or blended observational data. While deep learning models like RNNs with attention can model non-linear time-series effectively, they often lack integration with astrophysical constraints, limiting their physical reliability and interpretability [1], [2]. Furthermore, most existing approaches rely heavily on simulated datasets rather than large-scale real observations and rarely provide transparency into which parts of the light curve influence predictions. This gap highlights the need for a framework that combines the predictive power of deep learning, the transparency of attention-based interpretability, and the rigor of physics-informed constraints to analyze quasar variability using real observational data.

Machine learning (ML) has transformed the study of quasar variability, enabling the modeling of complex, non-linear light curves with improved predictive accuracy compared to traditional time-series analysis methods. While several studies have applied deep learning architectures such as recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and gated recurrent units (GRUs) to astronomical time-series, many have done so without ensuring the physical adequacy of the models in representing the underlying astrophysical processes driving variability. This oversight can lead to models that fit observational data well but lack alignment with established quasar physics. Furthermore, attention mechanisms capable of highlighting the most influential time segments for predictions have rarely been integrated with physics-informed constraints, missing an opportunity to produce results that are both interpretable and physically consistent [2]. Most existing approaches also rely heavily on simulated or small-scale datasets, with limited use of large, real observational quasar surveys, reducing their applicability to real-world astronomical analysis. Considering these research gaps, there is a pressing need for a framework that unifies attention-based interpretability, and real observational data to produce accurate, transparent, and meaningful predictions of quasar brightness variations.

2. RESEARCH OBJECTIVES

2.1 Main Objectives

The primary goal of this investigation is to create a new physics-informed, attention-enhanced recurrent neural network (RNN) framework that can effectively capture the complex, irregular variation in quasar luminosity based on large-scale, ground-based observational time-series datasets. This investigation attempts to reconcile purely data-driven astrophysical independent deep learning models with astrophysical grounded models by integrating Existing physical principles into the very architecture of the model. Doing so will not only improve the accuracy of the predictions made by the framework but also will guarantee that the output is always consistent with quasar physics as we know it. Additionally, this investigation attempts to solve the long-standing astrophysical interpretability challenge in applying machine learning models to astronomy by introducing attention mechanisms that identify the most significant parts of the quasar light curve. Having this dual focus on physical adherence and interpretability is intended to yield output that is not only scientifically reliable but is practically useful in terms of enabling us to gain insight into the dynamics of active galactic nuclei (AGN).

2.2 Specific Sub Objectives

The first sub-goal is to develop, deploy, and fine-tune RNN-based architecture for sequence modeling like Long Short-Term Memory (LSTM) networks with added attention mechanisms. Through this add-on, the network will selectively pay attention to the most knowledge-bearing parts of quasars' light curves, thus enhancing predictive accuracy as well as the interpretability of findings.

The second sub-objective is to incorporate physics-informed constraints such as light-travel time effects as well as energy conservation in the learning process in the model. These regulations are added in the loss function or architecture itself so that the predictions in the model are constrained to comply with existing astrophysical knowledge while still benefitting from the strength in deep learning in the detection of patterns.

The third sub-goal is to subject the proposed framework to intensive comparison with standard quasar variability analysis techniques, including popular statistical models like the damped random walk (DRW). This will involve verifying the correctness, robustness, and physical realism of model predictions compared to benchmark techniques to ensure that progress is not only statistically significant but also scientifically significant.

The fourth sub-goal is to train and test the model strictly on large-scale, true observational quasar datasets drawn from surveys as well as other time-domain astronomy initiatives. To steer clear of dependance on simulated or synthetic datasets, the study will guarantee that the model is fine-tuned to perform optimally in the real world

observational errors, e.g., noise, irregular sampling, incompleteness in light curves.

The fifth sub-goal is to offer a fine-grain interpretability investigation in the form of the outputs of attention to determine which specific time ranges and patterns of variability contribute most to the predictions in the model. Such studies will facilitate deducing if the model's reasoning is in line with astrophysical behavior that is understood as well as if the attention mechanism can provide new understanding of quasar variability that is not obvious to standard analysis approaches.

Lastly, the sixth sub-goal is to show that application of attention-based interpretability to physics-informed models can offer a generalized framework that can be extended to any other time-series challenge in astrophysics. Given that the methodology is framed as a framework instead of as the solution to a given defined problem, this study shall have broad impact to science in computational astrophysics, time-domain astronomy, as well as explainable artificial intelligence in science.

3. METHODOLOGY

This study adopts a multi-phase methodology that combines advanced sequence-modeling techniques with astrophysical constraints to produce accurate, interpretable models of quasar luminosity variability. The workflow integrates data preprocessing, model architecture design, physics-informed constraint embedding, training and evaluation, interpretability analysis, and validation against traditional approaches.

3.1 Requirements Gathering and Analysis

During this phase, all functional and non-functional needs of the intended system are discovered and investigated. For the quasar variability modeling project, this means defining scientific objectives (e.g., precise light curve variability modeling, inclusion of physics-informed constraints, and interpretability), the needs for data (e.g., type, format, and quality of quasar observational datasets), and the needs for computations (e.g., GPU utilization in the training of deep learning). Stakeholder feedback, including comments from astronomers, data scientists, and domain authorities are gathered to ensure that the system design is acceptable from both research objectives as well as practical use. This phase also recognizes the current literature to know the current state of knowledge as well as to clearly define distinct gaps that the project needs to fill.

3.2 Feasibility Study

The feasibility study evaluates whether the proposed research project can be successfully executed within the given constraints, such as time, resources, technology, and skills. It assesses the practicality of the approach and identifies potential risks early on. The study considers schedule, technical, and resource feasibility, ensuring that the project is both achievable and sustainable.

3.2.1 Schedule feasibility

This helps to determine if the project is feasible in the available time. For this research, this means approximating the time spent on data acquisition, preprocessing, formulation of model, training, validation, interpretability study, and validation. It also accounts for availability of datasets, time spent on computational experiments, time to interpret the output as well as to prepare the thesis. A reasonable schedule is formulated to ensure that all milestones will be achievable without any compromise in terms of quality.

3.2.2 Technical feasibility

The intended research will incorporate deep learning, machine learning, adversarial learning, approaches to advance the construction of an advanced data science platform

This evaluates if technical resources and know-how needed for the project exist. For this study, this involves checking the availability of quasar observational datasets, sufficiency in computing equipment (e.g., high-performance GPUs when building models), and appropriateness in software tools like Python, PyTorch, TensorFlow,This is in addition to checking the technical know-how in deep learning, attention mechanisms, and astro-data analysis. Technical feasibility is thus about being sure that the needed infrastructure and technical skills are available to effectively apply the proposed solution.

3.3 Tools and Technologies

The proposed research will employ a combination of programming languages, frameworks, and computational platforms to implement, train, and evaluate the physics-informed, attention-augmented deep learning models for quasar variability analysis.

- Python Language
- TensorFlow
- Libraries (AstroPy, SciPy, NumPy)
- Visual Studio Code
- Google Colab
- Jupyter Notebook

Note:The technologies listed here represent the primary proposed tools for the project. The final implementation may also incorporate additional technologies as needed

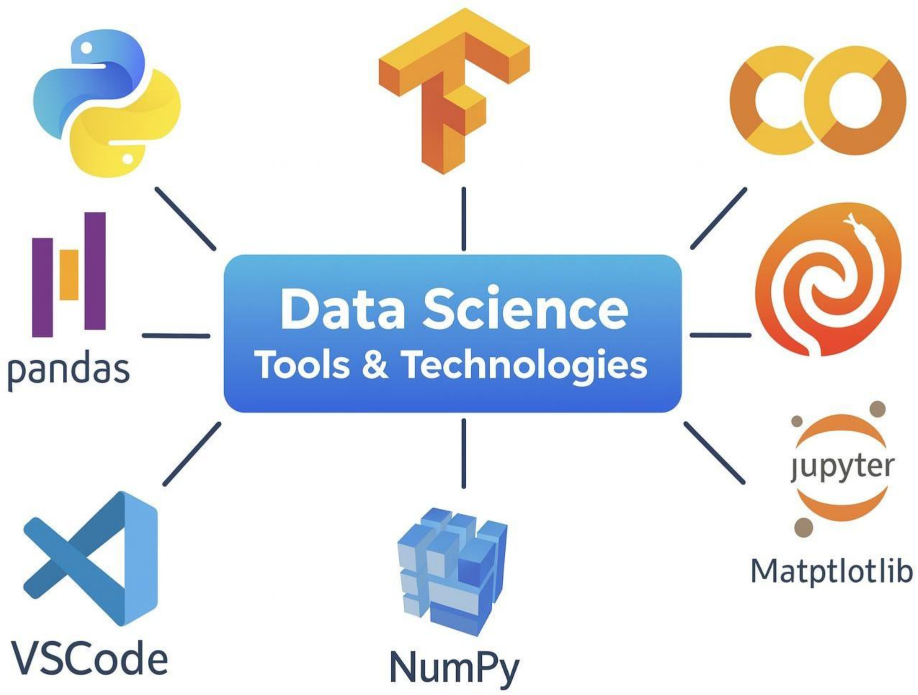

Figure 3 - Tools and Technologies

3.4 Overall System Diagram

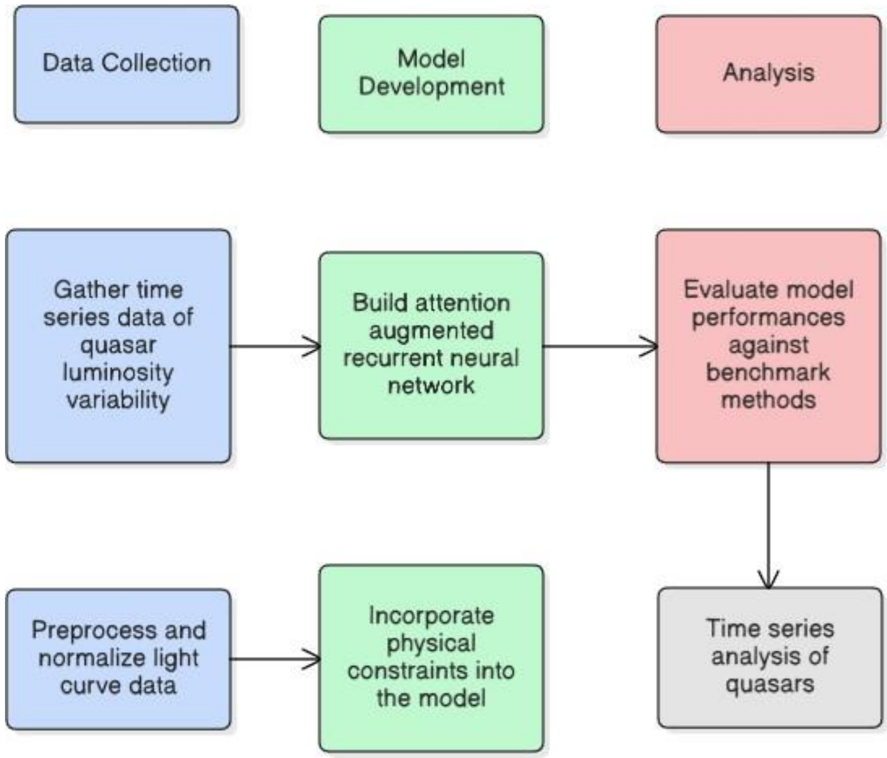

Figure 4 - System Diagram

- **Data Collection**

  - Gather time-series data of quasar luminosity variability:

  This step involves obtaining multi-epoch light curve observations of quasars from astronomical surveys

  - Preprocess and normalize light curve data:

  Before training, the raw light curve data is cleaned (removing outliers, handling missing values), normalized, and formatted for sequence modeling. This step ensures consistent scales and prepares data for the model input.

- **Model Development**

  - Build an attention-augmented recurrent neural network (RNN):

  The core model is an RNN architecture with an attention mechanism that allows the network to focus on the most informative parts of the quasar light curve when making predictions.

- Incorporate physical constraints into the model:

Physics-informed rules are embedded into the learning process. This ensures that predictions are not only accurate but also physically consistent with known quasar behavior.

- **Analysis**

  - Evaluate model performance against benchmark methods:

The trained model's predictions are compared with results from traditional variability analysis techniques (e.g., damped random walk models) to assess improvements in accuracy, interpretability, and physical reliability.

- **Outcome**

The output is a scientifically reliable, interpretable model that explains quasar brightness variations while adhering to astrophysical principles, offering both predictive accuracy and meaningful physical insights.

3.5 Component Diagram

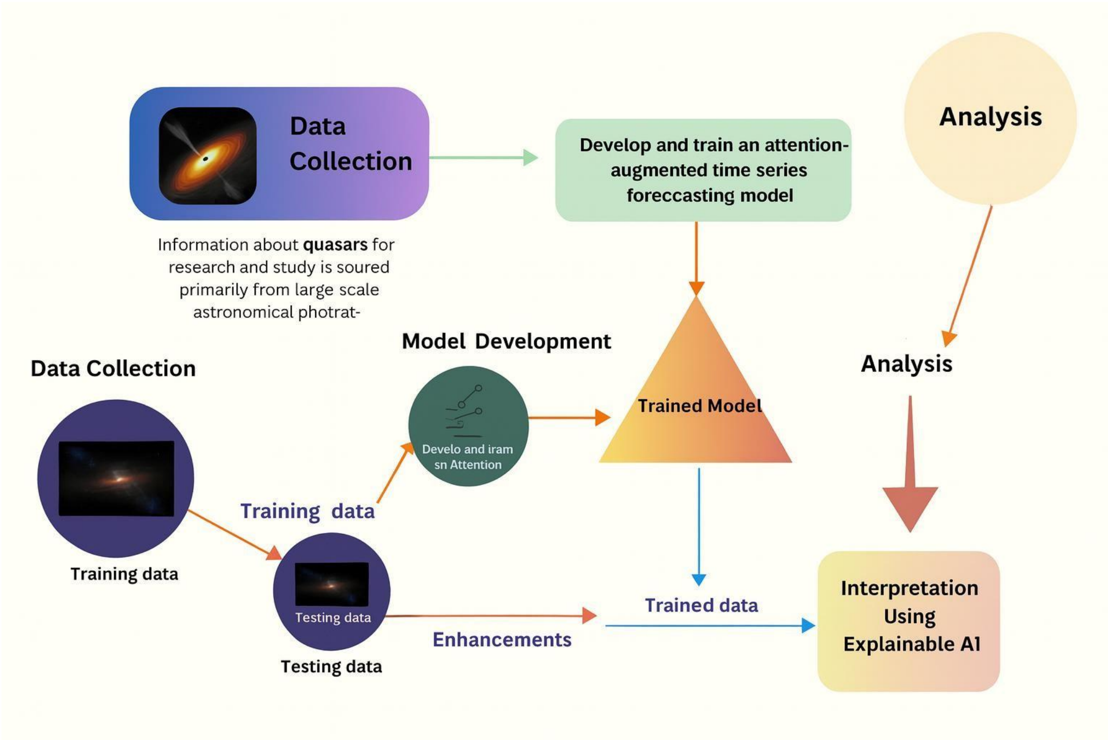

Figure 5 - Component Diagram

The Quasars Time Series Analysis component diagram shows the entire workflow from data acquisition to interpretation. High-volume surveys in astronomy produce quasar light curves that are divided into training as well as testing sets. Training data are used to train an attention-enhanced time-series predictive model that includes physics-informed constraints like light-travel time impacts as well as energy conservation to maintain physical plausibility. Once trained, the model is tested against the test dataset, with tuning added where necessary to optimize accuracy as well as eliminate bias. To evaluate, we compare our model's predictions with traditional methods and then confirm accuracy by checking how well it matches the actual observed light curves.

3.6 Project Requirements

3.6.1 Functional Requirements

The functional needs for the Time Series Analysis of Quasars system are to read in multiband quasar light curve data from sky surveys, to check the format for validity, and to carry out preprocessing like filling in missing values, normalizing across filters, and time-step alignment. The system should accommodate exploration analysis through plot output and statistical summary output and incorporate both classical baseline models like the damped random walk (DRW) as well as state-of-the-art deep learning architectures like RNNs, LSTMs, or GRUs with embedded attention mechanisms. Physics-informed constraints like light travel time effects as well as energy conservation should be included in the run to enforce physical plausibility. Hyperparameter tuning, model performance metrics like RMSE, MAE, and R^2, as well as vigorous comparison to baseline approaches should all be permitted in the framework. Interpretation features like attention heatmaps as well as output in terms of feature attribution to determine the most important regions in the light curves should be implemented. Other features are cross-validation, detection of biases, reproducibility through versioning in experiments, as well as automatic report writing with exportable output. Scalability through use of GPU acceleration should also be available in the system with a user-friendly interface to run the workflow, error treatment as well as logging, as well as extensive documentation to aid researchers.

3.6.2 Non-Functional Requirements

- Computational Efficiency & Performance: The system must utilize CPU/GPU resources efficiently, supporting mixed precision training and mini batching to minimize wall clock time and memory footprint. Typical training runs should be completed within predefined time budgets, and inference on a single multiband light curve should meet latency targets suitable for batch processing.
- Scalability: The architecture should scale horizontally to larger datasets (more quasars, longer light curves, more bands) and vertically to deeper models without significant performance degradation. Data pipelines and model training must handle

distributed execution (multi GPU/cluster) with near linear throughput gains where feasible.

- Data Accuracy, Quality & Reliability: Data ingestion must enforce schema checks (time stamps, bands, units), propagate measurement uncertainties, and track quality flags. Preprocessing steps (outlier handling, interpolation/masking, normalization) must be numerically stable and auditable to preserve scientific fidelity
- Fault Tolerance: The system should gracefully handle missing bands, irregular sampling, gaps, and noisy measurements. Failures in any stage (I/O errors, GPU OOM, convergence issues) must surface clear diagnostics and recover via checkpointing and resumable jobs.

3.7 Validation

In this research, validating the developed quasar variability models against real observational data is essential to ensure their accuracy and reliability. By comparing model predictions with actual light curves from large-scale astronomical surveys, the process evaluates how well the framework captures the true physical behavior of quasars. This validation not only strengthens confidence in the model's predictive capabilities but also ensures that the results are grounded in real-world astrophysical phenomena, thereby enhancing the scientific credibility and integrity of the research.

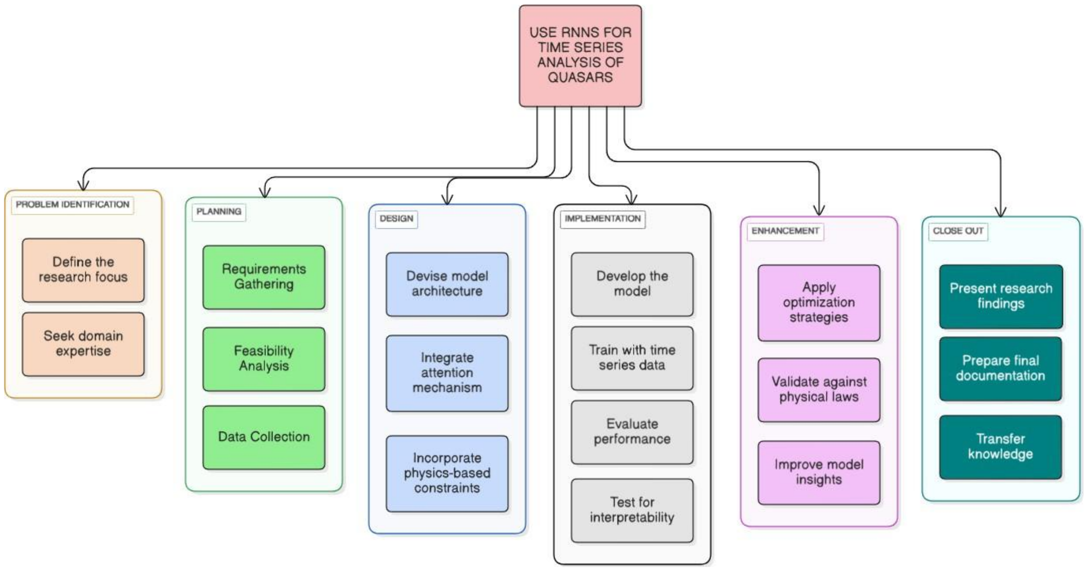

Figure 6 - work breakdown structure

The workflow is organized into six main phases: initiation, planning, design, implementation, enhancement, and close-out. The initiation phase involves defining the research scope, building the initial framework, and preparing the project proposal. In the planning phase, requirements are analyzed, feasibility studies are conducted, and relevant astronomical time-series datasets are collected from trusted observational sources. The design phase focuses on creating data architecture, structuring preprocessing pipelines, and designing the RNN model for time series analysis of quasar light curves. During the implementation phase, data preprocessing and model training are carried out, followed by optimization strategies to improve performance. The enhancement phase integrates explainable AI to interpret model predictions and aligns the outputs with physical theories for validation. Finally, the close-out phase consolidates the results through research publications, presentations, knowledge transfer, and recommendations for future work, ensuring the research is thoroughly documented and scientifically valuable.

3.8 Iterative Development

This is a Gantt chart (figure 7) that presents the schedule for the 28-week research in Time Series Analysis of Quasars. The first nine weeks are spent on brainstorming, registration, writing proposal, presentation, and final submission. Weeks 7 to 22 are mainly used in model implementation while progress reports are provided in Weeks 12 and 22. After writing a research paper and final report in the succeeding weeks, final presentation and viva are held in Week 26. The logbook, documentation, website, and final deployment are submitted in Weeks 27 and 28, effectively finishing the project.

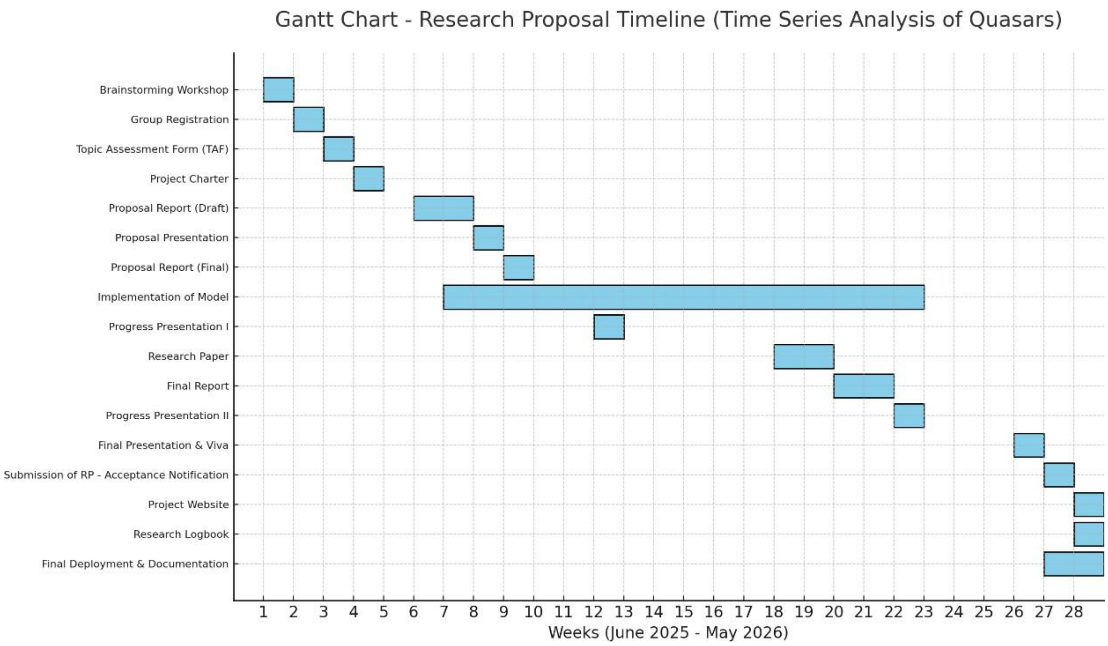

Figure 7 - Iterative Development

4. COMMERCIALIZATION AND CONTRIBUTION TO THE DOMAIN

The foreseen study on time series analysis of quasars through RNN can provide worthwhile expertise to space research and exploration entities like NASA, ESA, and large astronomical observatories. Using sophisticated deep learning models on quasar light curves, the research can enhance the detection, classification, and interpretation of complicated variation patterns in active galactic nuclei. Through this functionality, scientists can fine-tune cosmological models, predict masses for black holes, as well as investigate the physics of accretion processes through cosmic timescales. Additionally, through the inclusion of explainable AI, predictions are both scientifically interpretable as well as compliant with astrophysical theory, thus empowering astronomers to realize deeper knowledge on high-redshift quasars as well as the early cosmos. Through these advances, not only is the academic community improved, but practical use in terms of optimizing observational strategy as well as maximizing scientific return through telescope time is also achieved.

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